1.48 Measures of Central Tendency and Variation

Worked Solutions

1)

Set A:

Mean = 13

Median = 12

Mode = 3

Range = 25

Lower Quartile = 6

Upper Quartile = 20

Interquartile Range = 14

Set B:

Mean = 13

Median = 12

Mode = 3

Range = 25

Lower Quartile = 6

Upper Quartile = 20

Interquartile Range = 14

The statistics are the same for both sets of data despite coming from different data.

2 a)

The mean uses all values in its calculation and is good for roughly symmetric data.

2 b)

The median is unaffected by extreme values and is good for skewed data.

3 a)

Mean = 10.9 to 3sf

Standard Deviation = 6.10 to 3sf

3 b)

$\bar{x} - 2\sigma = 10.90909\ldots - 2 \times 6.09701\ldots = -1.28$ to 3sf

$\bar{x} + 2\sigma = 10.90909\ldots + 2 \times 6.09701\ldots = 23.1$ to 3sf

$\therefore$ 25 is an outlier

3 c)

Median = 10

Lower Quartile = 8, Upper Quartile = 15

Interquartile Range = $15 - 8 = 7$

3 d)

LQ $-1.5\times$ IQR $= 8 - 1.5 \times 7 = -2.5$

UQ $+1.5\times$ IQR $= 15 + 1.5 \times 7 = 25.5$

$\therefore$ There are no outliers

4 a)

Mean $= 20.5$

Standard Deviation $= 13.3$ to 3sf

4 b)

$\bar{x} - 2\sigma = 20.5 - 2 \times 13.28219\ldots = -6.06$ to 3sf

$\bar{x} + 2\sigma = 20.5 + 2 \times 13.28219\ldots = 47.1$ to 3sf

$\therefore 55$ is an outlier

4 c)

Median $= 19$

Lower Quartile $= 9.5$, Upper Quartile $= 27$

Interquartile Range $= 27 - 9.5 = 17.5$

4 d)

LQ $-1.5\times$ IQR $= 9.5 - 1.5 \times 17.5 = -16.75$

UQ $+1.5\times$ IQR $= 27 + 1.5 \times 17.5 = 53.25$

$\therefore 55$ is an outlier

5 a)

$\bar{x} = 45.8$ to 3sf

5 b)

If 59 was removed, the value of $\sum x$ would decrease, hence $\bar{x}$ would decrease.

6 a)

$\bar{x} = 56.5$

6 b)

If 40 is changed to 56.5, the value of $\sum x$ would increase, hence $\bar{x}$ would increase.

7 a)

$\bar{x} = \frac{\sum x}{n} = \frac{860}{5} = 172$

7 b)

$\bar{x} = \frac{\sum x}{n} = \frac{86753}{35} = 2478.657\ldots = 2480$ to 3sf

8)

$\bar{x} = \frac{\sum x}{n} \Rightarrow 52 = \frac{51324}{n} \Rightarrow n = \frac{51324}{52} = 987$

9)

$\sigma = \sqrt{\frac{56905}{10} - \left(\frac{613}{10}\right)^2} = 44.0$ to 3sf

10)

$\sigma = \sqrt{\frac{7322}{25} - \left(\frac{412}{25}\right)^2} = 4.61$ to 3sf

11)

$\sigma^2 = \frac{183075}{60} - \left(\frac{3305}{60}\right)^2 = 17.1$ to 3sf

12)

$\sigma^2 = \frac{212182}{9} - \left(\frac{1354}{9}\right)^2 = 942$ to 3sf

13 a)

$\sigma = 5.15$ to 3sf or $s = 5.65$ to 3sf

13 b)

If 125 was removed, then the data is less spread out, hence the standard deviation would decrease.

14 a)

$\sigma = 1.99$ to 3sf or $s = 2.18$ to 3sf

14 b)

If 43.6 was changed to 48.6, then the data is more spread out, hence the standard deviation would increase.

15 a)

$\bar{x} = 14.5$

15 b)

$\sigma = 4.35$ to 3sf or $s = 4.76$ to 3sf

15 c)

e.g. $x = 14$ and $y = 15$

x and y must be the same distance from the mean to keep the mean the same (so the mean of x and y must be 14.5), but less than 1 standard deviation away from the mean to reduce the standard deviation.

16 a)

$\bar{x} = 32$

16 b)

Median = 33

16 c)

e.g. $x = 31$ and $y = 33$

x and y must be the same distance from the mean to keep the mean the same (so the mean of x and y must be 32), but y must be less than 34 to reduce the median.

17 a)

$x = 22$

17 b)

$\frac{8+12+15+x+22+28}{6} = 17.05 \Rightarrow x = 17.3$

17 c)

$\frac{15+x}{2} = 16.3 \Rightarrow x = 17.6$

18 a)

$\bar{x} = 137.8333\ldots = 138$ cm to 3sf

18 b)

$s = 29.24764\ldots = 29.2$ cm to 3sf

18 c)

$\bar{x} - 2s = 79.33\ldots = 79.3$ to 3sf

$\bar{x} + 2s = 196.32\ldots = 196$ to 3sf

$\therefore$ 196 is an outlier

19)

$\bar{x} = 1.975$

Sample standard deviation is 1.46 to 3sf

(Standard deviation is 1.45 to 3sf)

20)

Mean is 1.68 to 3sf

Sample standard deviation is 0.704 to 3sf

(Standard deviation is 0.700 to 3sf)

21 a)

$\bar{x} = 2.6$

21 b)

Median = 2.5

22 a)

$\bar{x} = 22.75$

22 b)

Sample standard deviation is 4.50 to 3sf

(Standard deviation is 4.49 to 3sf)

22 c)

Using the midpoints assumes that the data is evenly distributed within the class intervals.

23 a)

11.5

23 b)

$\bar{x} = 10.4$ kg to 3sf

23 c)

Sample standard deviation is 2.25 kg to 3sf

(Standard deviation is 2.22 kg to 3sf)

24 a)

27

24 b)

66 (as of writing) is England's state retirement age

24 c)

$\bar{x} = 38.9$ years to 3sf

Sample standard deviation is 10.8 years to 3sf

(Standard deviation is 10.7 years to 3sf)

25 a)

$\bar{x} = £46,000$ to 3sf

25 b)

Sample standard deviation is £8,200 to 3sf

(Standard deviation is £7,950 to 3sf)

25 c)

These are only estimates as we have used the midpoints of the class intervals. This assumes that the raw data is evenly distributed within the class intervals, which may not actually be the case.

25 d)

This data only comes from one publishing house, so is only a very small sample. Also, the salaries in London would be expected to be higher than other places in the UK.

26)

$\bar{x} = \frac{\sum x}{n} \Rightarrow 9 = \frac{\sum x}{33} \Rightarrow \sum x = 297$

$\sigma^2 = \frac{\sum x^2}{n} - \bar{x}^2 \Rightarrow 2 = \frac{\sum x^2}{33} - 9^2 \Rightarrow \sum x^2 = 2739$

27)

$\bar{x} = \frac{\sum x}{n} \Rightarrow 90.2 = \frac{\sum x}{100} \Rightarrow \sum x = 9020$

$\sigma^2 = \frac{\sum x^2}{n} - \bar{x}^2 \Rightarrow 10.96 = \frac{\sum x^2}{100} - 90.2^2 \Rightarrow \sum x^2 = 814700$

28)

Any hours worked over 8 hours must be discounted, so the two times that she worked over 8 hours should be counted as precisely 8 hours.

i.e.

Hours Worked (t)	Frequency
$0 \leq t < 2$	12
$2 \leq t < 4$	17
$4 \leq t < 6$	21
$6 \leq t < 8$	7
$t = 8$	2

$\bar{t} = 3.949153\ldots$

Ebba's mean gross pay is $3.949153\ldots \times £15.60 = £61.61$ to the nearest penny

29)

$1 = \sqrt{\frac{80000}{n} - \left(\frac{40000}{n}\right)^2} \Rightarrow 1 = \frac{80000}{n} - \frac{1600000000}{n^2}$

$\Rightarrow n^2 - 80000n + 1600000000 = 0$

$\Rightarrow (n - 40000)^2 = 0 \Rightarrow n = 40000$

30)

$32 = \frac{\sum x}{8} \Rightarrow \sum x = 256$

$30 \times 8 = 240$ so if k is the value of the piece of data that is removed, it must be replaced by $k + 16$. One standard deviation either side of the mean is $(30 - 8.70, 30 + 8.70) = (21.3, 38.7)$, so

$k + 16 < 38.7 \Rightarrow k < 22.7$

So the two pieces of data that can be replaced are 19 and 21. If 19 is removed it must be replaced with 35. If 21 is removed it must be replaced with 37.

31)

$\bar{x} = \frac{\sum x}{n} \Rightarrow k = \frac{\sum x}{10} \Rightarrow \sum x = 10k$

$$\sigma = \sqrt{\frac{\sum x^2}{n} - \bar{x}^2} \Rightarrow 5 = \sqrt{\frac{\sum x^2}{10} - k^2}$$

$$\Rightarrow 25 = \frac{\sum x^2}{10} - k^2 \Rightarrow \sum x^2 = 250 + 10k^2$$

Two more pieces of data are added:

$\sum x = 10k + (k-1) + (k+1) = 12k$

$$\begin{aligned}\sum x^2 &= 250 + 10k^2 + (k-1)^2 + (k+1)^2 \\ &= 250 + 10k^2 + k^2 - 2k + 1 + k^2 + 2k + 1 = 12k^2 + 252\end{aligned}$$

$\therefore \sigma = \sqrt{\frac{12k^2+252}{12} - \left(\frac{12k}{12}\right)^2} = \sqrt{k^2 + 21 - k^2} = \sqrt{21}$

32)

$\bar{x} = \frac{\sum x}{n} \Rightarrow 200 = \frac{\sum x}{10} \Rightarrow \sum x = 2000$

$$\sigma = \sqrt{\frac{\sum x^2}{n} - \bar{x}^2} \Rightarrow k = \sqrt{\frac{\sum x^2}{10} - 200^2}$$

$$\Rightarrow k^2 = \frac{\sum x^2}{10} - 40000 \Rightarrow \sum x^2 = 400000 + 10k^2$$

Two more pieces of data are added:

$\sum x = 2000 + (2+k) + (2-k) = 2004$

$$\begin{aligned}\sum x^2 &= 400000 + 10k^2 + (2+k)^2 + (2-k)^2 \\ &= 400000 + 10k^2 + k^2 + 4k + 4 + k^2 - 4k + 4 = 12k^2 + 400008\end{aligned}$$

$\therefore 99 = \sqrt{\frac{12k^2+400008}{12} - \left(\frac{2004}{12}\right)^2}$

$\Rightarrow 9801 = k^2 + 33334 - 27889 \Rightarrow k^2 = 4356 \Rightarrow k = 66$

$k \neq -66$ as the standard deviation can only be greater than or equal to zero.

33)

$\bar{x} = \frac{\sum x}{n} \Rightarrow 30 = \frac{\sum x}{23} \Rightarrow \sum x = 690$

$$\sigma = \sqrt{\frac{\sum x^2}{n} - \bar{x}^2} \Rightarrow 5 = \sqrt{\frac{\sum x^2}{23} - 30^2}$$

$$\Rightarrow 25 = \frac{\sum x^2}{23} - 900 \Rightarrow \sum x^2 = 21275$$

One more student sits the test:

$\sum x = 690 + 35 = 725$

$\sum x^2 = 21275 + 35^2 = 22500$

$\therefore \bar{x} = \frac{725}{24} = 30.2$ to 3sf

$\sigma = \sqrt{\frac{22500}{24} - \left(\frac{725}{24}\right)^2} = 4.996$ to 4sf

34)

$\bar{x} = \frac{\sum x}{n} \Rightarrow 40 = \frac{\sum x}{25} \Rightarrow \sum x = 1000$

$$\sigma = \sqrt{\frac{\sum x^2}{n} - \bar{x}^2} \Rightarrow 8 = \sqrt{\frac{\sum x^2}{25} - 40^2}$$

$$\Rightarrow 64 = \frac{\sum x^2}{25} - 1600 \Rightarrow \sum x^2 = 41600$$

One student is removed from the test:

$\sum x = 1000 - 52 = 948$

$\sum x^2 = 41600 - 52^2 = 38896$

$\therefore \bar{x} = \frac{948}{24} = 39.5$ to 3sf

$\sigma = \sqrt{\frac{38896}{24} - \left(\frac{948}{24}\right)^2} = 7.77$ to 3sf

35)

y	5	$\frac{45}{2}$	40	75	115

36)

$\bar{x} = 85,\ \sigma_x = 12$

$y = 10x$

$\Rightarrow \bar{y} = 10 \times 85 = 850$

$\sigma_y = 10 \times 12 = 120$

37)

$\bar{x} = 120,\ \sigma_x = 5$

$y = 3x + 50$

$\Rightarrow \bar{y} = 3 \times 120 + 50 = 410$

$\sigma_y = 3 \times 5 = 15$

38)

$\bar{y} = 500,\ \sigma_y = 120$

$y = 8x - 10 \Rightarrow x = \frac{y+10}{8}$

$\Rightarrow \bar{x} = \frac{500+10}{8} = 63.75$

$\sigma_x = \frac{120}{8} = 15$

39)

$\bar{y} = 250,\ \sigma_y = 45$

$y = \frac{20-3x}{9} \Rightarrow x = \frac{20-9y}{3}$

$\Rightarrow \bar{x} = \frac{20-9\times250}{3} = -\frac{2230}{3} = -743$ to 3sf

$\sigma_x = \frac{9\times45}{3} = 135$

40 a)

Mean is $\bar{x} = 25.42857143\ldots = 25.4°\text{C}$ to 3sf

Sample standard deviation is $s_x = 2.070196678\ldots = 2.07°\text{C}$ to 3sf

(Standard deviation is $\sigma_x = 1.916629695\ldots = 1.92°\text{C}$ to 3sf)

40 b)

Mean is $\bar{y} = \frac{9\times25.42857143...}{5} + 32 = 77.8°\text{F}$ to 3sf

Sample standard deviation:

$s_y = \frac{9\times2.070196678...}{5} = 3.73°\text{F}$ to 3sf

(Standard deviation:

$\sigma_y = \frac{9\times1.916629695...}{5} = 3.45°\text{F}$ to 3sf)

41 a)

Mean is $\bar{x} = 61.42857143\ldots°\text{F}$

Sample standard deviation is $s_x = 1.988059595\ldots = 1.99°\text{F}$ to 3sf

(Standard deviation $\sigma_x = 1.840585532\ldots = 1.84°\text{F}$ to 3sf)

41 b)

Mean is $\bar{y} = \frac{5(61.42857143...-32)}{9} = 16.3°\text{C}$ to 3sf

Sample standard deviation is $s_y = \frac{5\times1.988059595...}{9} = 1.10°\text{C}$ to 3sf

(Standard deviation is $s_y = \frac{5\times1.840585532...}{9} = 1.02°\text{C}$ to 3sf)

42)

Mid-range $= \frac{75+28}{2} = 51.5$

43 a)

Mode $= 1$

43 b)

Mid-range $= \frac{6+0}{2} = 3$

43 c)

Median $= 1$

44)

x	Frequency	Cumulative Frequency
$0 \leq x < 5$	35	35
$5 \leq x < 10$	60	95
$10 \leq x < 15$	15	110
$15 \leq x \leq 20$	10	120

Median, Q_2, position is $\frac{120}{2} = 60^{\text{th}}$, which is in the $5 \leq x < 10$ class interval.

$60 - 35 = 25$

$Q_2 = 5 + \frac{25}{60}(10 - 5) = \frac{85}{12} = 7.08$ to 3sf

Alternative Method:

$\frac{Q_2-5}{60-35} = \frac{10-5}{95-35} \Rightarrow Q_2 = \frac{85}{12} = 7.08$ to 3sf

45)

x	Frequency	Cumulative Frequency
$0 \leq x < 8$	82	82
$8 \leq x < 12$	74	156
$12 \leq x < 20$	62	218
$20 \leq x \leq 24$	44	262

Median, Q_2, position is $\frac{262}{2} = 131^{\text{st}}$, which is in the $8 \leq x < 12$ class interval.

$131 - 82 = 49$

$Q_2 = 8 + \frac{49}{74} \times (12 - 8) = \frac{394}{37} = 10.6$ to 3sf

Alternative Method:

$\frac{Q_2 - 8}{131 - 82} = \frac{12 - 8}{156 - 82} \Rightarrow Q_2 = \frac{394}{37} = 10.6$ to 3sf

46)

x	Frequency	Cumulative Frequency
$10 \leq x < 30$	55	55
$30 \leq x < 70$	95	150
$70 \leq x < 80$	150	300
$80 \leq x < 110$	75	375
$110 \leq x \leq 140$	25	400

Median, Q_2, position is $\frac{400}{2} = 200^{\text{th}}$, which is in the $70 \leq x < 80$ class interval.

$200 - 150 = 50$

$Q_2 = 70 + \frac{50}{150}(80 - 70) = \frac{220}{3} = 73.3$ to 3sf

Alternative Method:

$\frac{Q_2 - 70}{200 - 150} = \frac{80 - 70}{300 - 150} \Rightarrow Q_2 = \frac{220}{3} = 73.3$ to 3sf

47)

x	Frequency	Cumulative Frequency
$50 \leq x < 60$	12	12
$60 \leq x < 70$	16	28
$70 \leq x < 80$	28	56
$80 \leq x < 100$	32	88
$100 \leq x \leq 120$	14	102

Lower Quartile, Q_1, position is $\frac{102}{4} = 25.5^{\text{th}}$, which is in the $60 \leq x < 70$ class interval.

$25.5 - 12 = 13.5$

$Q_1 = 60 + \frac{13.5}{16}(70 - 60) = \frac{1095}{16} = 68.4375$

Alternative Method:

$\frac{Q_1 - 60}{25.5 - 12} = \frac{70 - 60}{28 - 12} \Rightarrow Q_1 = \frac{1095}{16} = 68.4375$

Upper Quartile, Q_3, position is $\frac{3 \times 102}{4} = 76.5^{\text{th}}$, which is in the $80 \leq x < 100$ class interval.

$76.5 - 56 = 20.5$

$Q_3 = 80 + \frac{20.5}{32}(100 - 80) = \frac{1485}{16} = 92.8125$

Alternative Method:

$\frac{Q_3 - 80}{76.5 - 56} = \frac{100 - 80}{88 - 56} \Rightarrow Q_3 = \frac{1485}{16} = 92.8125$

$IQR = 92.8125 - 68.4375 = \frac{195}{8} = 24.375$

48 a)

x	Frequency	Cumulative Frequency
$0 \leq x < 15$	48	48
$15 \leq x < 40$	93	141
$40 \leq x < 55$	117	258
$55 \leq x < 70$	152	410
$70 \leq x \leq 95$	146	556
$95 \leq x < 110$	121	677
$110 \leq x < 135$	94	771
$135 \leq x < 150$	29	800

20th percentile, A, position is $\frac{20}{100} \times 800 = 160^{\text{th}}$, which is in the $40 \leq x < 55$ class interval.

$160 - 141 = 19$

$A = 40 + \frac{19}{117}(55 - 40) = \frac{1655}{39} = 42.4$ to 3sf

Alternative Method:

$\frac{A - 40}{160 - 141} = \frac{55 - 40}{258 - 141} \Rightarrow A = \frac{1655}{39} = 42.4$ to 3sf

80th percentile, B, position is $\frac{80}{100} \times 800 = 640^{\text{th}}$, which is in the $95 \leq x < 110$ class interval.

$640 - 556 = 84$

$B = 95 + \frac{84}{121}(110 - 95) = \frac{12755}{121} = 105$ to 3sf

Alternative Method:

$\frac{B - 95}{640 - 556} = \frac{110 - 95}{677 - 556} \Rightarrow B = \frac{12755}{121} = 105$ to 3sf

20th to 80th percentile range $= 105.41322\ldots - 42.43589\ldots = 62.97732\ldots = 63.0$ to 3sf

48 b)

$\frac{55 - 42.43589\ldots}{55 - 40} \times 117 + 152 + 146 + \frac{105.41322\ldots - 95}{110 - 95} \times 121$

$= 98 + 152 + 146 + 84 = 480$

Alternative Method:

60% of 800 is 480

49)

x (pH)	Frequency	Cumulative Frequency
$6 \leq x < 6.2$	12	12
$6.2 \leq x < 6.4$	27	39
$6.4 \leq x < 6.6$	38	77
$6.6 \leq x < 6.8$	22	99
$6.8 \leq x < 7.0$	9	108

Upper Quartile, Q_3, position is $\frac{3\times 108}{4} = 81^{\text{st}}$, which is in the $6.6 \leq x < 6.8$ class interval.

$81 - 77 = 4$

$Q_3 = 6.6 + \frac{4}{22}(6.8 - 6.6) = \frac{73}{11} = 6.64$ to 3sf

Alternative Method:

$\frac{Q_3 - 6.6}{81-77} = \frac{6.8-6.6}{99-77} \Rightarrow Q_3 = \frac{73}{11} = 6.64$ to 3sf

50)

Height (m)	Frequency	Cumulative Frequency
13 to 15	7	7
16 to 18	15	22
19 to 21	23	45
22 to 24	9	54
25 to 26	2	56

Lower Quartile, Q_1, position is $\frac{56}{4} = 14^{\text{th}}$, which is in the 16 to 18 class interval.

$14 - 7 = 7$

$Q_1 = 15.5 + \frac{7}{15}(18.5 - 15.5) = \frac{169}{10} = 16.9$

Alternative Method:

$\frac{Q_1 - 15.5}{14-7} = \frac{18.5-15.5}{22-7} \Rightarrow Q_1 = \frac{169}{10} = 16.9$

51)

Length (cm)	Frequency	Cumulative Frequency
6 to 7	56	56
8 to 9	84	140
10 to 11	129	269
12 to 13	76	345
14 to 15	43	388

30th percentile, Q, position is $\frac{3\times 388}{10} = 116.4^{\text{th}}$, which is in the 8 to 9 class interval.

$116.4 - 56 = 60.4$

$Q = 7.5 + \frac{60.4}{84}(9.5 - 7.5) = \frac{1877}{210} = 8.94$ to 3sf

Alternative Method:

$\frac{Q - 7.5}{116.4-56} = \frac{9.5-7.5}{140-56} \Rightarrow Q = \frac{1877}{210} = 8.94$ to 3sf

So we expect 70% of the fish to be longer than 8.94 cm.

52)

Age (years)	Frequency	Cumulative Frequency
25 to 30	9	9
31 to 40	32	41
41 to 50	19	60
51 to 65	26	86
66 to 80	12	98

Median, Q_2, position is $\frac{98}{2} = 49^{\text{th}}$, which is in the 41 to 50 class interval.

$49 - 41 = 8$

$Q_2 = 41 + \frac{8}{19}(51 - 41) = \frac{859}{19} = 45.2$ to 3sf

$\frac{Q_2 - 41}{49 - 41} = \frac{51 - 41}{60 - 41} \Rightarrow Q_2 = \frac{859}{19} = 45.2$ to 3sf